

Aaditya Khanuja

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EDUCATION

University of California San Diego

La Jolla, CA

Bachelor of Science in Computer Science | IDEA Scholar

Sep. 2024 – Jun. 2028

- **Relevant Coursework:** Data Structures & Algorithms in Java, C Systems Programming & Software Tools, Logic Design of Digital Systems, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Econometrics
- **Activities and Awards:** Computer Science and Engineering Society, Triton Quantitative Trading, Hedge Fund Club, SDx UCSD, Triton Droids, Blackstone LaunchPad, Division-1 Men's Tennis (walk-on), Provost Honors (2x)

EXPERIENCE

Quantitative Methods Researcher

Sep. 2024 – Present

Triton Quantitative Trading

La Jolla, CA

- Engineered Python/C++ algorithmic strategies backtested on 8K+ observations, improving Sharpe ratio to 0.76
- Built a low-latency C++ trading pipeline that reduced slippage and increased simulated returns by 32%
- Built Python data pipeline extracting real-time financial API data, processing 50MB+ daily for backtesting

Consumers & Healthcare Sector Lead

Sep. 2024 – Present

Hedge Fund Club

La Jolla, CA

- Built Python/R DCF models to analyze 50 equities, generating 14% annual return across \$1M simulated portfolio
- Directed 10-member team producing 25+ research reports and automated database pipeline in Python
- Developed R screening algorithm using stochastic forecasting, identifying 8 positions with potential upside

AI Research Intern

Jun. 2024 – Aug. 2024

Senslytics Corporation

Remote

- Preprocessed 30,000+ samples in Python (pandas/NumPy); built ETL pipeline processing 200MB+ daily data
- Developed Python (PyTorch) NLP tool reducing alignment errors by 56% for Shell oil-rig operations
- Contributed to CorroX ILI Alignment Tool, improving data processing and cutting analysis time by 33%

Strategy Intern

Feb. 2024 – May. 2024

Catalyst Business Partners

Remote

- Conducted due diligence for 4 clients identifying 15+ opportunities; quantified \$50K savings using Excel models
- Supported Oracle Fusion implementation automating processes for 20% faster close and \$70K annual savings
- Built Excel cost-benefit model with scenario analysis; presented NPV and payback projections to C-suite

PROJECTS

Multimodal Sentiment Analysis for THIA | Python, PyTorch, NumPy, HuggingFace

May. 2025 – Present

- Designed multimodal sentiment analysis model using Python (PyTorch/HuggingFace transformers) processing video/audio streams for therapeutic AI agent, achieving 77% classification accuracy with 200ms latency
- Optimized model evaluation framework across 20K+ training samples from diverse demographics; implemented GPU-accelerated inference pipeline processing with real-time parallelized preprocessing, reducing latency by 85ms
- Engineered end-to-end pipeline with signal processing and late-fusion neural networks leveraging NumPy for tensor operations; accepted to Blackstone LaunchPad, San Diego Tech Conference, and Student AI Research Summit

Fan Engagement Model | Python, Random Forest, XGBoost, scikit-learn, Seaborn

Feb. 2025 – Mar. 2025

- Placed 3rd nationally in 2025 Milwaukee Bucks Hackathon (30+ competing universities); built Python Random Forest and XGBoost ensemble models processing 100,000+ fan records achieving 81% churn prediction accuracy
- Designed Seaborn visualizations presenting crossover event analysis (projected 15% revenue increase), referral program ROI of 22%, and solutions for boosting fan engagement to Milwaukee Bucks and Modine executives
- Collaborated with 5-student team using GitHub with structured branching and code reviews; developed modular Python codebase following PEP 8 standards with comprehensive docstrings and detailed technical documentation

SKILLS AND INTERESTS

Languages: Python, Java, C, C++, PostgreSQL, MATLAB, R, LaTeX

Frameworks & Libraries: PyTorch, NumPy, pandas, scikit-learn, SpecAugment, Chroma, XGBoost, Seaborn

Business Tools: Excel, Word, PowerPoint, Google Suite, Oracle Fusion

Developer Tools: VS Code, GitHub, IntelliJ IDEA, Jupyter Notebook, Google Colab, Overleaf

Interests: Tennis, Cycling, Golf, Painting, Piano, Reading, Volunteering, Astronomy